EXECUTIVE SUMMARY

Diagnosing Revenue Concentration and Delivery Failures in a Global Distribution Network

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Strong overall financial performance does not necessarily indicate a healthy business. Despite maintaining a positive profit margin, DataCo faces significant operational inefficiencies and commercial dependencies that may threaten its long-term performance.

This analysis examines where revenue and profit are concentrated, the extent of loss-making activity, delivery performance, discount practices, and market contributions to identify the areas requiring management attention.

Business Objective

The objective of this project was to identify the commercial and operational factors driving and limiting DataCo's performance and translate analytical findings into practical recommendations for protecting profitability, improving delivery reliability, and strengthening commercial resilience.

Methodology

The project followed a structured business intelligence workflow:

Business Understanding → Data Audit → Data Preparation → Data Modeling → Exploratory Analysis → Business Insights → Recommendations → Power BI Dashboard

Using SQL Server, I cleaned and validated 180,519 order-line records and conducted exploratory analysis across commercial, financial, geographic, and operational dimensions. A star-schema data model and a four-page Power BI dashboard were developed to support business performance monitoring and decision-making.

Key Findings

- **Loss-making activity is hidden behind positive overall profitability**

Despite an overall profit margin of 10.78%, approximately 21.15% of analyzed records generate losses. This indicates that aggregate profitability conceals a substantial segment of underperforming activity, potentially weakening the company's financial performance if left unaddressed.

- **Revenue concentration creates commercial exposure**

Fan Shop contributes approximately 47% of company sales and profit, while the Fishing category contributes approximately 19%. Although these are major contributors to business performance, such concentration increases exposure to changes in demand or deterioration in the performance of these segments.

- **Most sales and profit come from low and medium discount levels**

Low and medium discount groups account for approximately 83% of sales and 84% of profit. These groups represent the largest share of commercial activity, but their contribution alone does not establish whether current discount practices are optimal. High-discount transactions therefore require closer profitability assessment to avoid unnecessary margin erosion.

- **Europe leads commercial performance**

Europe generates the highest sales and profit among the five analyzed markets. However, higher revenue contribution does not automatically indicate the highest investment potential. Comparing market profitability and growth opportunities is necessary to ensure that resources are allocated effectively.

Priority Recommendations

- **Improve delivery reliability**

Prioritize fulfillment coordination and delivery planning across high-volume shipping methods, focusing corrective measures on markets with the highest delay rates. This would reduce delivery failures, improve customer experience, and help protect customer retention.

- **Protect transaction profitability**

Introduce targeted profitability controls for underperforming transactions by reviewing shipping costs, discounts, product mix, and destination markets. This would help prevent unprofitable sales from eroding overall profitability.

- **Diversify revenue sources**

Develop promising departments and categories beyond Fan Shop and Fishing while maintaining the performance of leading segments. This would reduce commercial dependency and strengthen the company's resilience to changes in demand.

- **Strengthen discount discipline**

Preserve disciplined pricing while reassessing high-discount transactions that may weaken margins. This would help protect profitability without unnecessarily restricting discounting as a commercial tool.

- **Allocate resources toward profitable markets**

Evaluate market opportunities based on profitability and growth potential rather than revenue alone. This would support more effective resource allocation and direct commercial investment toward sustainable growth opportunities.

Business Impact

The analysis indicates that DataCo's strongest opportunities lie in operational improvement rather than revenue generation alone. Addressing delivery delays, reducing loss-making transactions, and mitigating revenue concentration would strengthen profitability while reducing commercial risk.

At a Glance

Metric	Result
Sales (2015–2018)	$36.78M
Profit Margin	10.78%
Late Deliveries	54.8%
Loss-making Orders	21.15%
Largest Revenue Driver	Fan Shop (~47%)
Markets	5
Countries	164
Orders	65,752
Customers	20,652
Tools	SQL Server, Power BI

Full analysis, SQL scripts, Power BI dashboard, and documentation:
https://github.com/FARAHBENSALEM01/DataCo-Commercial-Distribution-Analysis/tree/main